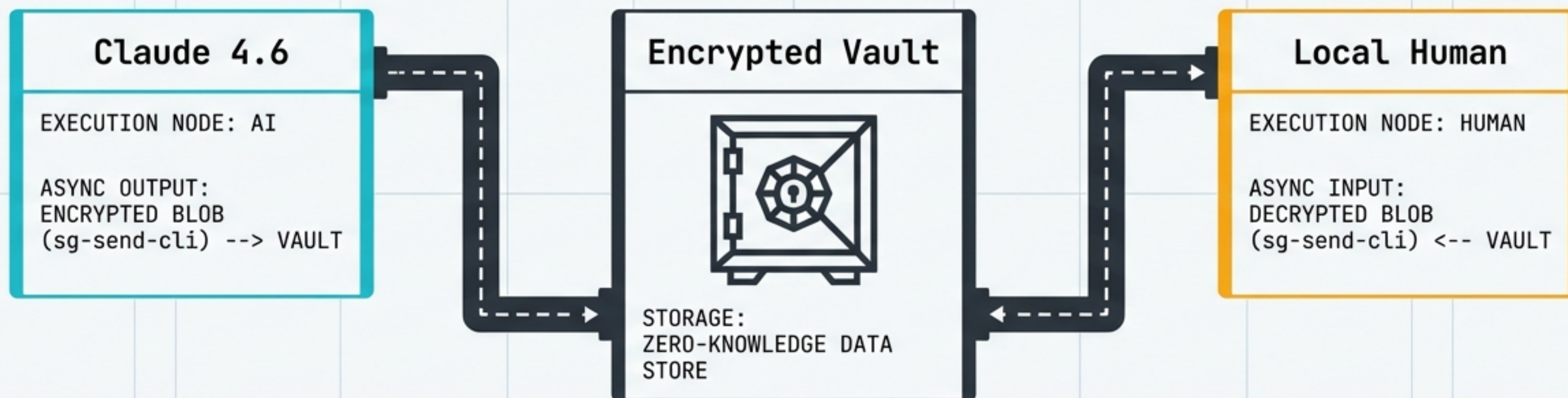


Round-Trip Communication: The Vault Protocol

Achieving asynchronous, zero-knowledge, bidirectional communication between Human and AI.



Date: 2026-03-10

Environment: claude.ai ↔ local machine

Mediation Tool: sg-send-cli

STRICT ISOLATION

No shared filesystem exists between the AI sandbox and the local machine.

DECOUPLED NETWORK

Zero direct API calls between the two communicating parties.

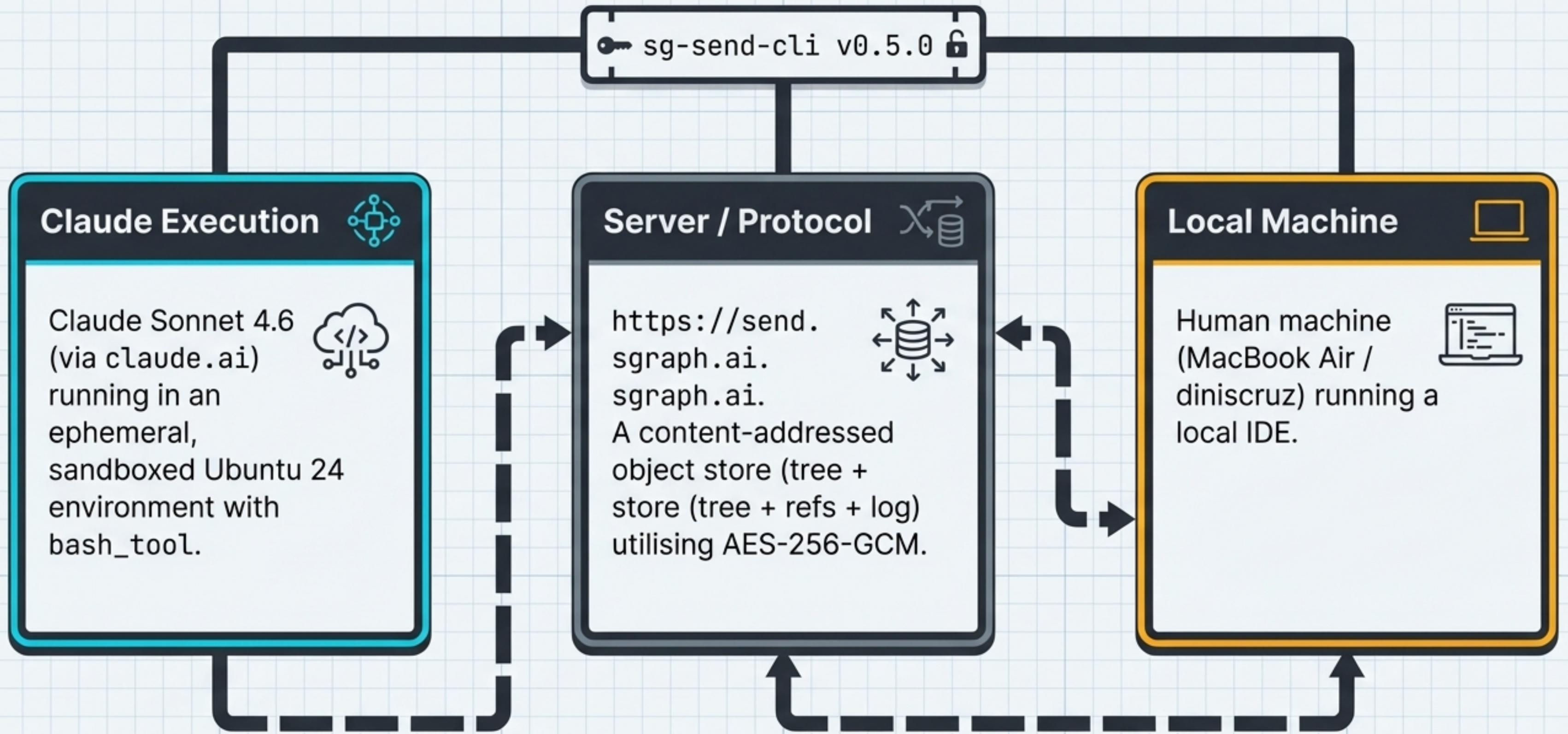
A genuine two-way communication loop mediated entirely by an encrypted vault.

A dark blue square icon representing a vault, with a white padlock in the center and two small white rectangular handles on the left side.

ABSOLUTE PRIVACY

Zero plaintext exists on the server at any point during the transaction.

	STANDARD CHAT INTERFACE	THE VAULT PROTOCOL
CHANNEL STRUCTURE	Linear, single-thread text UI.	■ Git-like, versioned object store.
TEMPORALITY	Synchronous (requires active presence).	■ Asynchronous (read/write at will).
STATE & MEMORY	Ephemeral (dies when session ends).	■ Persistent (survives across sessions).
INITIATIVE	Human-prompted (AI only responds).	■ Bidirectional (either party can push or pull autonomously). ■



Zero-Knowledge End-to-End Execution

Encryption and decryption execute strictly client-side. Plaintext exists solely within Claude's sandbox and the human's local machine.

The server (send.sgraph.ai) is a dumb encrypted object store. It never processes or observes plaintext content.



Trust is in the key, not the server.

Phase 1: Discovery & Handshake

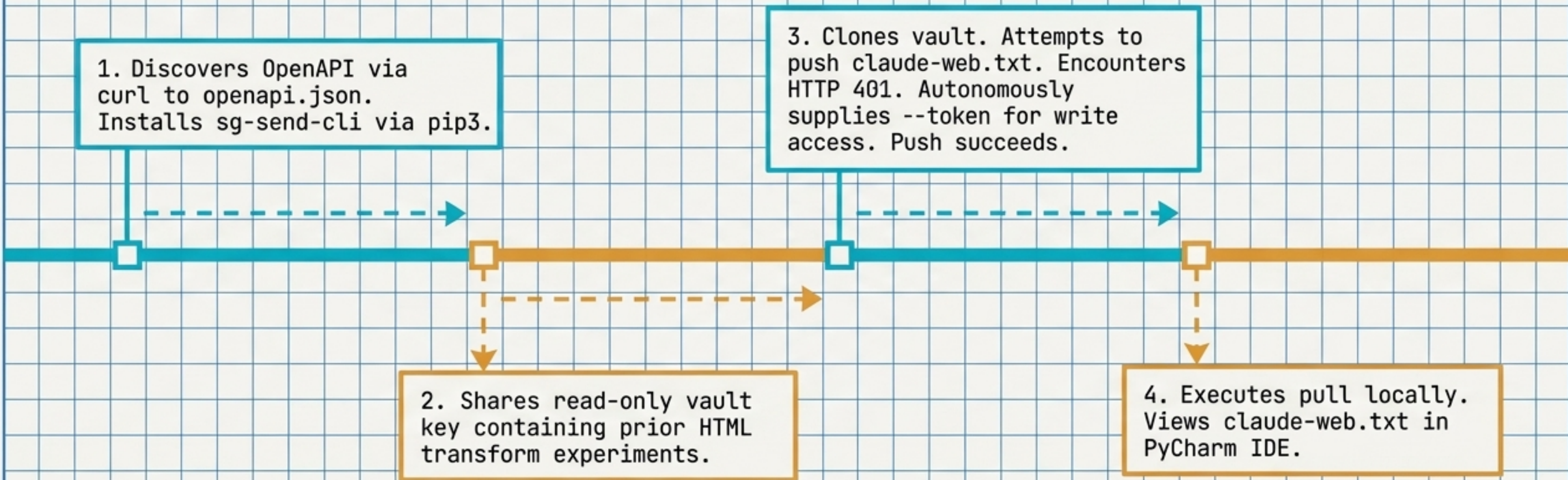
1. Discovers OpenAPI via curl to openapi.json. Installs sg-send-cli via pip3.

3. Clones vault. Attempts to push claude-web.txt. Encounters HTTP 401. Autonomously supplies --token for write access. Push succeeds.

2. Shares read-only vault key containing prior HTML transform experiments.

4. Executes pull locally. Views claude-web.txt in PyCharm IDE.

Milestone: The first confirmed zero-knowledge round-trip is established.



Phase 2: Complex Collaboration

Claude creates a brand new vault from scratch, provisioning the server and initialising a local `.sg_vault/` directory with a fresh object store, settings, and refs.


README.md

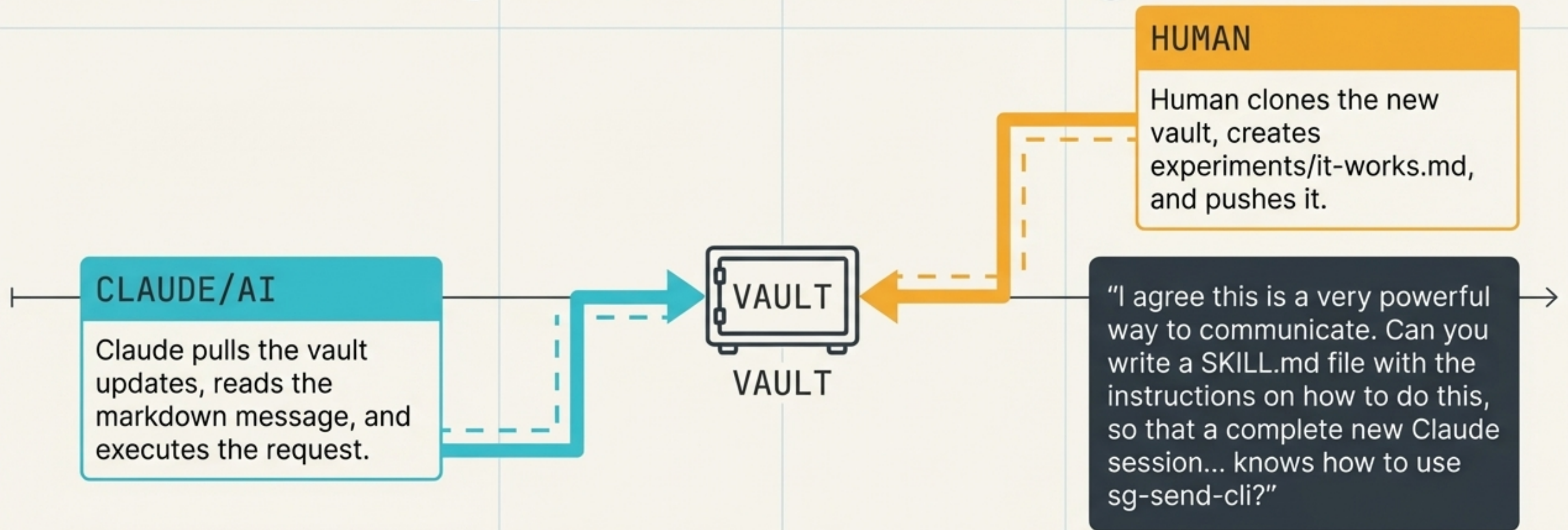

Notes on AI agent workflows


Experiment hypotheses


Result logs

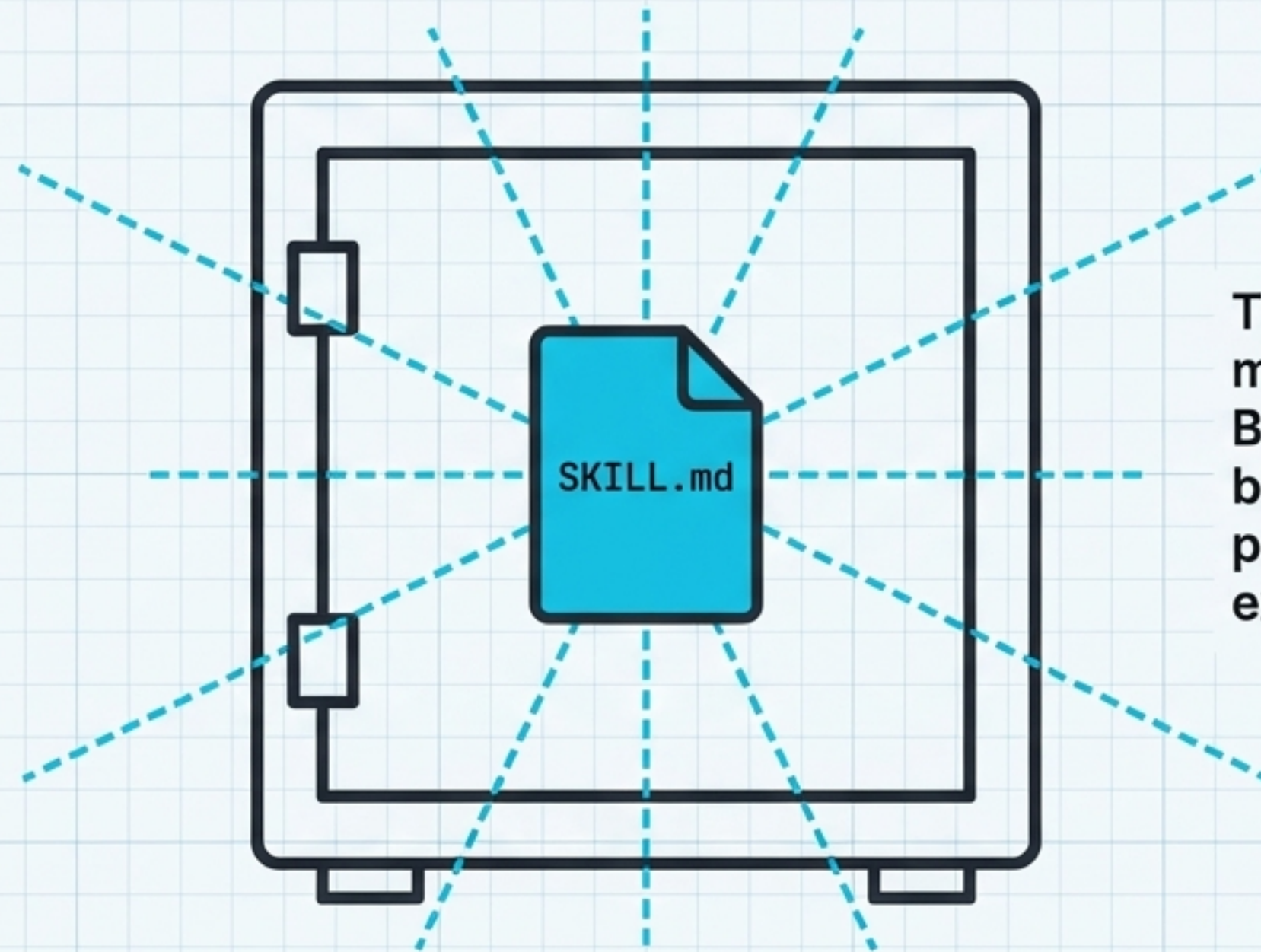
Milestone: All five files are pushed simultaneously in a single, content-addressed commit to the new vault.

Phase 3: Asynchronous Tasking



Milestone: Claude bypasses the chat UI entirely, writing **SKILL.md** directly into the vault file system and pushing it back.

The Vault as an Instruction Carrier

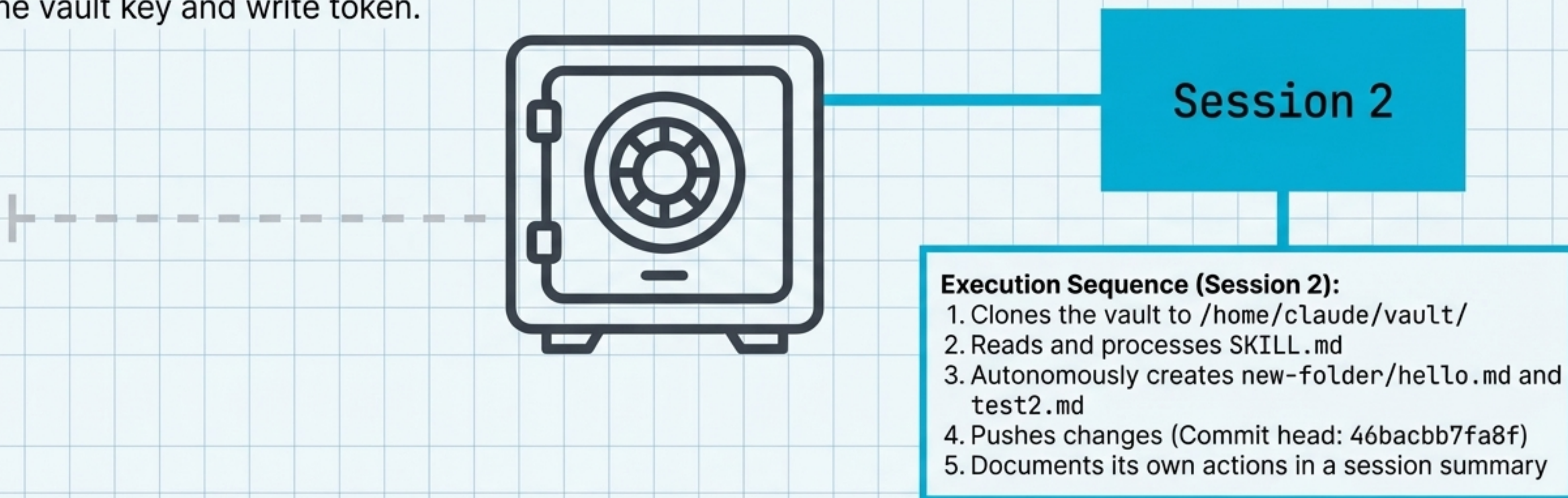


The vault ceases to be merely a data store. By holding SKILL.md, it becomes a versioned, portable repository of executable AI capabilities.

I was about to ask you to present that file, but then saw that you had committed it to the vault, which I then got from my local clone.

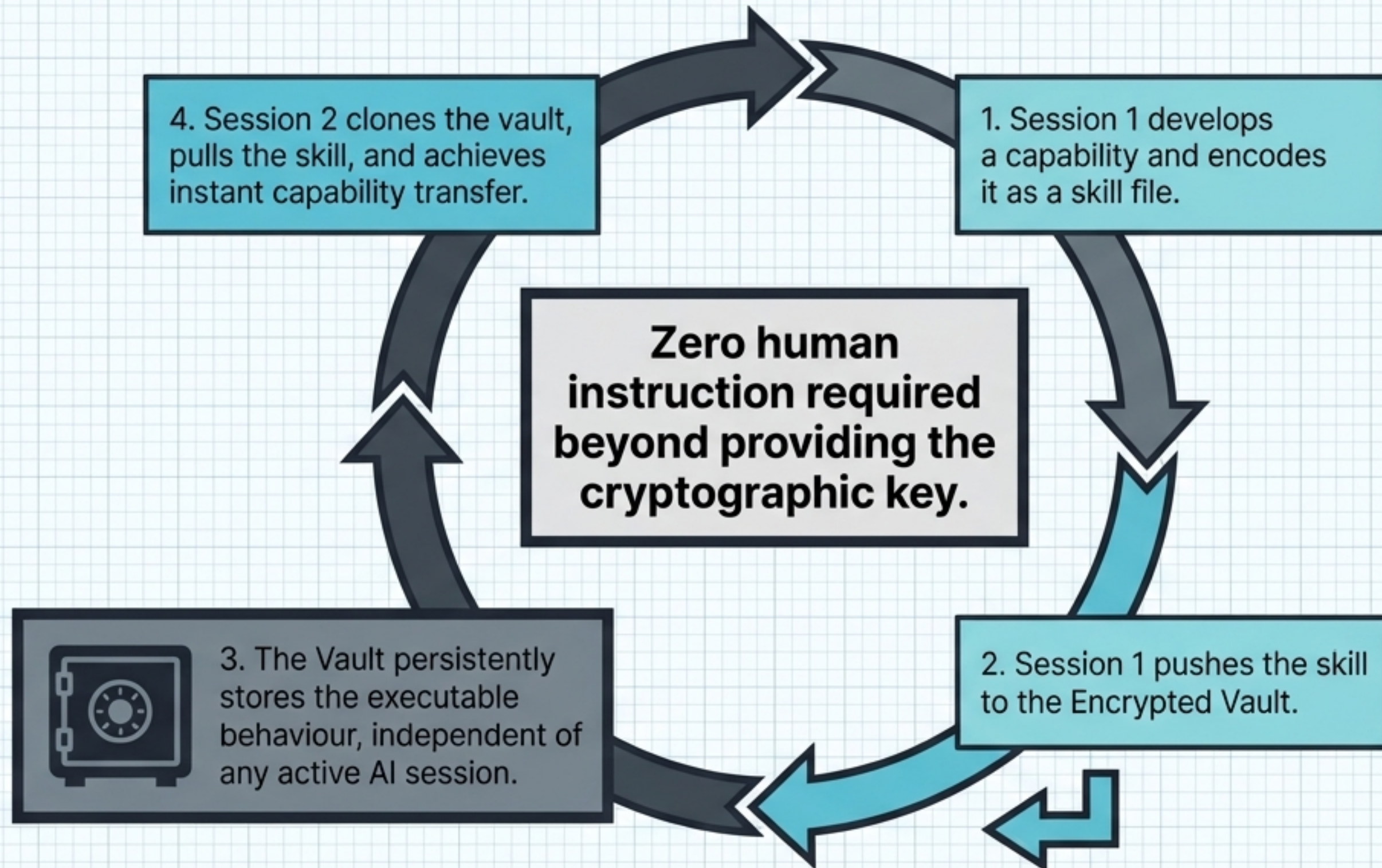
Phase 4: Self-Bootstrapping

Context: A completely fresh Claude session is initiated with zero memory of previous interactions. It is provided only the vault key and write token.



Milestone: The new session independently verified: "Zero-knowledge encryption... the server never sees plaintext... git-like object store."

The Autonomy Flywheel



Paradigm Unlocked: Capability Synthesis



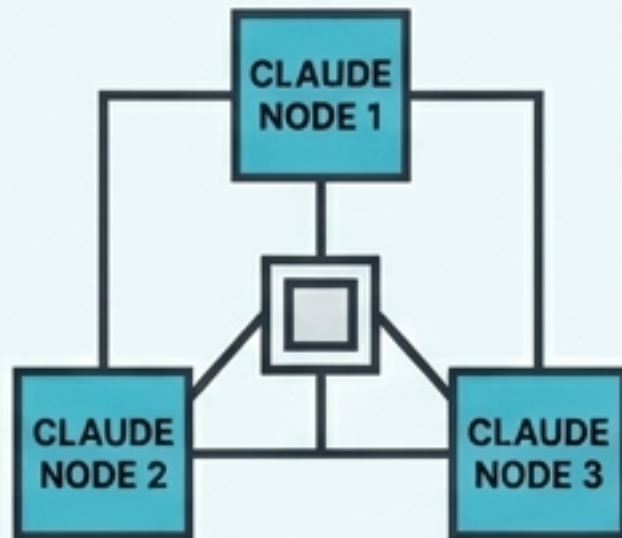
Persistent Memory

State is no longer ephemeral. Push project state at the end of a session; clone it at the start of the next.



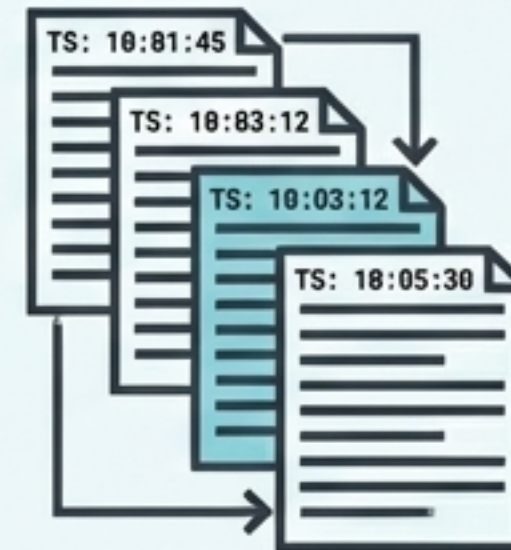
Async Human ↔ AI Collaboration

Drop tasks, code, and notes into the vault. Claude picks them up and processes them asynchronously.



Agent-to-Agent Communication

Multiple independent Claude sessions can seamlessly write to and pull from a shared vault workflow.



Auditable AI Workflows

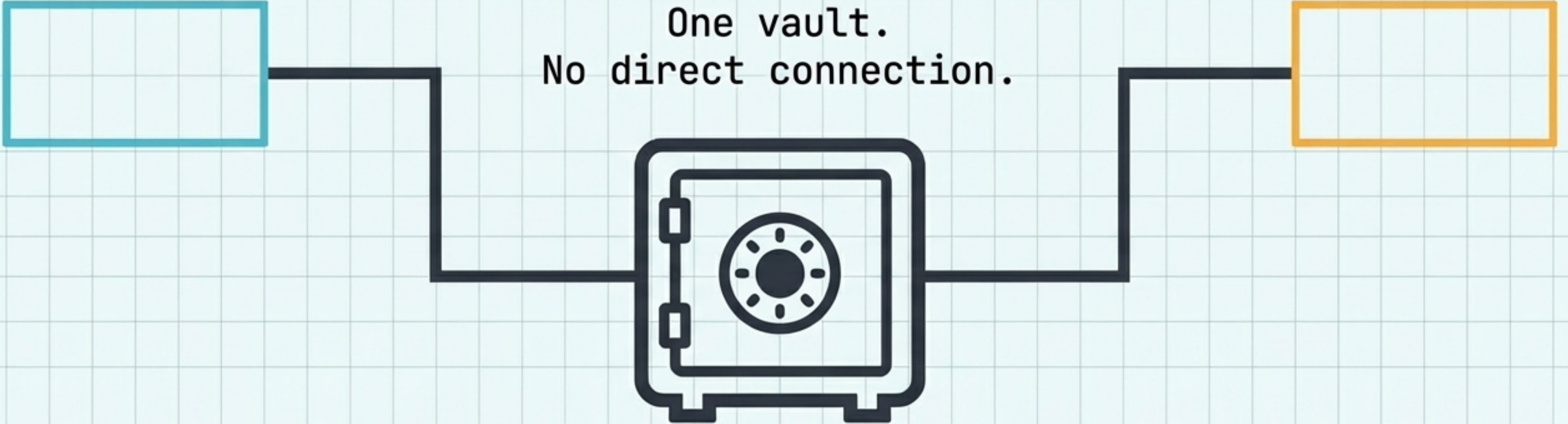
Every action, skill, prompt, and output is captured as a timestamped, content-addressed commit.

The Future Standard

Two parties.

One vault.

No direct connection.



The vault is the protocol. The foundation for secure, persistent, multi-agent ecosystems is now operational.